Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · AI-Powered Products · LLMs in Production · ML Pipelines · Evaluation Harnesses · Python · AWS · AI-Native

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SUMMARY

Senior software engineer with 9+ years building production AI systems, ML pipelines, and large-scale data processing end-to-end. Currently at Intuit building LLM-powered features with Anthropic Claude - RAG pipelines, structured outputs, evaluation harnesses with accuracy targets, and guardrails that keep AI outputs reliable in production. Built **AgentProbe** - an open-source LLM evaluation and accuracy tracking platform with a 5-metric eval engine, CI regression gating, and cost-aware multi-provider routing. Built **CrashClarity** - an ML pipeline on 104K+ UK accident records using Random Forest and Logistic Regression (85%+ accuracy) with systematic accuracy tracking and model iteration. At Entrata (7+ years, 100K+ users), owned high-volume production data pipelines and backend systems in a regulated multi-tenant SaaS platform. AI-native engineering practice: daily use of Claude Code and AI agents for design, code, testing, and operations. Strong across Python, ML (scikit-learn, XGBoost), data pipelines, AWS, Docker, and Kubernetes. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

LLMs in Production · AI-Powered Products · Anthropic Claude | ML Pipelines · scikit-learn · XGBoost · Model Training

Evaluation Harnesses · Accuracy Targets · CI Regression | LLM-as-Judge · Golden Datasets · Accuracy Tracking

RAG Pipelines · Hybrid Search · Embeddings · Retrieval | Guardrails · Safety Controls · HITL · Production Reliability

Data Pipelines · High-Volume Processing · Low-Latency Systems | Measurement & Estimation Systems · Forecasting · ML Modeling

AI-Native Engineering · Claude Code · AI Agents Daily | Python · FastAPI · TypeScript · Java · Node.js | AWS (Lambda · S3 · RDS · Glue · SQS) · Docker · Kubernetes

PostgreSQL · Redis · Snowflake · SQLite | Design Docs · Technical Ownership · Mentoring · Code Review | GitHub Actions CI/CD · OpenTelemetry · Observability

EXPERIENCE

Software Engineer

Jun 2026 - Present

Intuit — Virginia, USA

- **LLMs in production with accuracy targets:** Integrate Anthropic Claude into Python/FastAPI financial services - RAG pipeline with retrieval grounding, prompt design for structured outputs, and evaluation harnesses (LLM-as-judge + golden datasets) with CI regression gates that catch quality drops before every release.
- **ML pipelines from training to production:** Build scikit-learn and XGBoost ML pipelines - data ingestion, feature engineering, model evaluation, and prediction APIs - with accuracy tracking and iterative improvement based on production outcomes.
- Containerize and deploy via Docker + GitHub Actions CI/CD to AWS with CloudWatch monitoring; develop React/TypeScript front ends; implement guardrails with input validation and fallbacks for production reliability.

Senior Software Engineer

Oct 2020 - Jul 2024

Entrata — Apartment & Affordable Housing SaaS (regulated multi-tenant, 100K+ monthly active users)

- **High-volume production data pipelines:** Owned the HUD income-limit compliance ETL pipeline for 6 years - ingesting federal source data and transforming it into ~180,000 normalized domain records with validation, reconciliation, and rollback each cycle. Zero compliance violations across the full ownership period.
- **Production systems at scale:** Built RabbitMQ async event-driven processing for rent updates across thousands of tenant records (idempotency, retry logic, failure routing); PHI-compliant AWS S3 migration (Libsodium encryption, 100% coverage, zero incidents); Docker microservices redesign (50% scalability improvement, 30% faster deployments).
- Optimized 200+ SQL queries (30% performance gain); Python/Bash automation eliminating 80% of manual workload; internal debugging tooling cutting triage time by 40%; mentored 5+ engineers; 300+ code reviews across 120+ Agile sprints.

Junior Software Engineer

Sep 2018 - Oct 2020

Entrata India — Pune

- Built backend modules (Python, PHP, Java, PostgreSQL); delivered large-scale data integration and reconciliation workflows (1.6M records); led AWS S3 cloud migration from legacy storage.

PROJECTS

AgentProbe - LLM Evaluation & Accuracy Tracking Platform 2025 | Python, FastAPI, PostgreSQL, CI/CD | github.com/venkatalakshmiparimala/AgentProbe

- Production eval platform: offline golden-dataset evals, LLM-as-judge scoring, online accuracy metrics over time, CI regression harness gating releases on accuracy thresholds. 5-metric eval engine (accuracy, hallucination rate, retrieval quality, completeness, compliance). Cost-aware routing across Anthropic, OpenAI, Azure OpenAI, AWS Bedrock, Groq, and Ollama. Full tracing and structured logging per run.

Leasel - Production AI Platform: LLMs · Agents · RAG · Guardrails 2026 | Python, LangGraph (8-agent), FastAPI, ChromaDB, CI/CD | github.com/venkatalakshmiparimala/Lease_complaince_rag

- End-to-end AI-powered product shipped to production: 8-agent LangGraph workflow, hybrid search RAG (OCR ingestion, TF-IDF + semantic reranking, citation grounding), HITL guardrails, and CI eval harness. 93.6% precision across 1,240 test cases - accuracy target set, measured, and hit.

CrashClarity - ML Modeling & Estimation (104K Records) 2024 | Python, scikit-learn, Random Forest, Logistic Regression, R | GitHub Pages

- ML pipeline on 104K+ UK road accident records - Random Forest and Logistic Regression models achieving 85%+ accuracy; systematic model evaluation, accuracy tracking, and iteration; interactive analytics dashboard. Pattern extraction from large structured datasets to predict and quantify outcomes.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia | GPA 3.87 May 2026

B.Tech Computer Science & Engineering — JNTU Anantapur 2017